

1 Agroforestry SmartAgroforest: Poplar biomass modelling

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3 Abstract

4 Short-rotation agroforestry systems are increasingly promoted as a low-carbon feedstock
5 option for bio-based value chains, but their economic viability depends on coordinated
6 decisions on stand establishment, multi-cycle harvesting, and product cascading in spa-
7 tially distributed supply chains. This paper develops a mixed-integer linear programming
8 (MILP) model for the integrated design and operation of an agroforestry biomass sup-
9 ply chain with explicit age-lag constraints that link consecutive harvests at each site.
10 The formulation couples allometric yield functions for different biomass types with a
11 multi-product, multi-period network design model including storage, processing, and
12 product quality cascades. The model is applied to a stylised case study of Central Euro-
13 pean poplar-based alley-cropping systems to explore trade-offs between harvest rotation
14 length, product portfolios, and infrastructure configuration. We outline a computational
15 experimentation plan to test the impact of allometric parameter uncertainty, price sce-
16 narios, and policy incentives on the optimal deployment of agroforestry systems in cleaner
17 bioeconomy pathways.

18 *Keywords:* Agroforestry, Short-rotation coppice, Biomass supply chain, Product
19 cascading

20 1. Introduction

21 Agroforestry systems that integrate trees with agricultural crops (silvoarable systems)
22 are increasingly recognised as a promising land-use option to reconcile climate mitigation,
23 biodiversity conservation, and rural income generation on the same hectare of land (Toens-
24 meier, 2017). Temperate agroforestry systems (AFS) based on fast-growing species such

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25 as poplar, willow, alder, and black locust can deliver substantial biomass yields while im-
26 proving soil quality and providing multiple ecosystem services compared to conventional
27 monocultures (Huber et al., 2018). Biomass produced in AFS is considered as a sustain-
28 able feedstock and can be used either energetically, chemically or materially (e.g. in the
29 building sector). The latter two options are generally preferred over the energetic usage
30 from a cleaner production perspective as biogenic carbon is stored in products during
31 their life span which contributes to climate change mitigation.

32 Despite substantial research on ecological advantages and pioneering initiatives to
33 establish AFS in practice, the share of AFS in agricultural production systems is still
34 small. E.g., in Germany about 1200 hectares of silvoarable AFS are established in 2025,
35 which accounts for about 0.02% of total agricultural land use (Deutscher Fachverband
36 für Agroforstwirtschaft (DeFAF) e.V., 2026). One of the core challenges in establishing
37 AFS systems at large scale in practice is a lack of opportunities to generate substantial
38 revenues from woody biomasses of AFS. On the other hand, the rise of bioeconomy creates
39 substantial demands for woody biomass and requires a reliable and cost-efficient feedstock
40 supply. As woody biomasses are bulky feedstocks with low energy density, the design
41 of cost-efficient supply chains is crucial to ensure the economic viability of AFS-based
42 value chains. This requires coordinated decisions on stand establishment, multi-cycle
43 harvesting, and product cascading in spatially distributed supply chains, which can be
44 supported by integrated optimisation models that link stand-level growth dynamics to
45 regional supply chain design.

46 In the last two decades, a large body of research has analysed biomass supply chains
47 for bioenergy and biorefineries using mathematical programming models, often in the
48 form of mixed-integer linear programming (MILP) (Zahraee et al., 2020; Sharma et al.,
49 2013). These models typically optimise network configuration, feedstock sourcing, and
50 logistics under cost or profit objectives. They often treat biomass availability as an exoge-
51 nous input or rely on simplified agricultural production decisions. Moreover, downstream
52 processing of biomasses is concentrated on one particular value chain e.g. bioenergy plants
53 consuming all produced biomasses. Recent work has started to integrate operational de-
54 tails such as various biomass types, biomass growth models and regeneration processes
55 in multi-period supply chain models.

56 The present paper contributes to this literature in three ways. First, we propose
57 a novel MILP formulation for AFS-based biomass supply chain design that explicitly
58 links establishment decisions and consecutive harvest decisions, ensuring that harvesting
59 decisions at each site are consistent with minimum and maximum rotation ages. Second,
60 we embed allometric yield functions for different biomass types generating age-specific
61 yields for different woody biomass types (like stems, branches, or leaves and twigs). Third,
62 these different woody biomass types can be processed in specific downstream industries,
63 like chemical or paper industry as well as energy sector. The model aims to provide
64 strategic decision support for establishing regional AFS-based bioeconomy ecosystems
65 connecting multiple industries and product markets.

66 The remainder of the paper is structured as follows. Section 2 reviews the relevant
67 literature on agroforestry biomass systems, allometric biomass modelling, and biomass
68 supply chain optimisation. Section 3 summarises the MILP formulation for the agro-
69 forestry supply chain with age-lag constraints. Section 4 presents the case study instan-
70 tiation, the base-case optimisation results, and the sensitivity analysis, examining the
71 implications for cleaner production strategies.

72 **2. Literature review**

73 Agroforestry biomass systems, stand-level growth models, and biomass supply chain
74 optimisation have each been studied extensively, but they are rarely combined in an inte-
75 grated framework that links age-dependent stand dynamics to regional product-cascading
76 value chains (Zahraee et al., 2020; Sharma et al., 2013; Espinoza et al., 2017). This sec-
77 tion reviews (i) agroforestry systems with emphasis on system types, economics, and the
78 cascading use of biomass as the primary revenue driver; (ii) biomass growth models at
79 tree and stand level with explicit attention to the allometric–diameter–stand-biomass
80 pathway; and (iii) biomass supply chain design using mathematical programming. The
81 section concludes by identifying the modelling gap addressed in this paper.

82 *2.1. Agroforestry systems*

83 Agroforestry deliberately integrates woody perennials with crops and/or livestock on
84 the same management unit, with the aim of enhancing productivity, profitability, and

85 environmental performance (Nair, 1993; Toensmeier, 2017). In the temperate zone, three
86 distinct system types can be distinguished based on the spatial arrangement of trees and
87 the rotation length at which woody biomass is harvested.

88 **Short-rotation coppice (SRC)** is defined as a monoculture plantation of fast-
89 growing woody species covering the entire management unit, with planting densities
90 of 8,000–20,000 cuttings $\cdot\text{ha}^{-1}$. After each harvest the root system regenerates without
91 replanting. Rotation cycles are typically 3–7~years and total stand lifetime is 20–25~years
92 (Trnka et al., 2008; Faasch and Patenaude, 2012). SRCs are no AFSs in the narrow sense,
93 but it is helpful to benchmark biomass production.

94 **Short-rotation agroforestry systems (SRAFS)** integrate SRC-type tree strips
95 (typically 1–4 rows of the same fast-growing species like poplar or willow) into an oth-
96 erwise arable field in an alley-cropping arrangement. Crop production continues in the
97 inter-row strips (Huber et al., 2018; Graves et al., 2007). Thus, tree strips typically cover
98 10–30% of the total parcel area. Rotation lengths and total stand lifetime in SRAFS are
99 comparable to SRCs. In SRAFS all trees are effectively edge trees, exposed to higher light
100 and water availability than interior SRC trees. This boundary effect raises individual-tree
101 biomass accumulation but also intensifies competition with adjacent crop rows (Huber
102 et al., 2018).

103 **Long-rotation agroforestry systems (LRAFS)** use the same alley-cropping spa-
104 tial structure as SRAFS but with rotation cycles of 12–15~years or longer (Graves et al.,
105 2007; Nair, 1993). At this rotation length, individual trees develop substantial stem di-
106 ameters and the proportion of high-value timber increases markedly, enabling premium
107 material uses (veneer, sawlogs, engineered wood products) that are inaccessible in short-
108 rotation systems. Therefore, often timber-quality tree species (such as walnut, cherry,
109 oak, or high-grade poplar clones) are planted.

110 SRAFS and LRAFS are not clearly differentiated in practice. First, a clear cut-off age
111 is hard to define and, more importantly, the intended use of AFS biomass is not always
112 clear and can also change over time. Nonetheless, when establishing AFS, the structure
113 of AFS and its intended use after harvest drives planting decisions and, subsequently,
114 economic viability. Thus, in the present paper SRAFS and LRAFS are not considered as
115 separate AFS systems, but the interplay of intended use, establishment, and harvesting

116 decisions is analysed.

117 2.2. *Economic rationale of landowners*

118 The decision of a landowner to establish an AFS involves weighing substantial up-
119 front costs and long-term opportunity costs against a diversified revenue stream. On the
120 cost side, establishment requires site preparation, planting material, and planting labour,
121 typically amounting to 1,500–3,500 €/ha for SRC or SRAFS (Faasch and Patenaude,
122 2012; Testa et al., 2014). Recurrent harvest and refitting costs (machine hire, chipping,
123 transport to roadside) add approximately 200–400 €/ha per rotation depending on stand
124 density and equipment availability (Testa et al., 2014; Spinelli et al., 2009). During
125 the rotation, maintenance costs of AFS arise on a yearly basis. Next to expenses for
126 establishment, maintenance and harvesting, opportunity costs arise as no revenues from
127 crops are generated on tree strips. In regions with fertile soils and high cereal prices,
128 yearly crop revenues may sum up to 400–600 €/ha (Faasch and Patenaude, 2012; Graves
129 et al., 2007). Thus, opportunity costs are often the dominant cost driver in SRAFS
130 profitability analyses.

131 On the **benefit side**, the primary revenue of AFS is the sale of harvested woody
132 biomass, whose magnitude depends critically on rotation length, species, and site pro-
133 ductivity. For poplar-based SRAFS in Central Europe, typical yields range from 10
134 to 25 t of dry matter biomass per ha for typical rotation lengths (Huber et al., 2018;
135 Niemczyk and Thomas, 2021). Additionally, positive crop-yield effects in the inter-row
136 strips of SRAFS and LRAFS may result. Tree rows act as windbreaks, reducing soil
137 erosion and evapotranspiration; they provide partial shading that moderates heat stress
138 during drought years; and their root systems may improve water retention in the top-
139 soil (Graves et al., 2007; Grünewald et al., 2007). Empirical measurements in Central
140 European alley-cropping trials indicate that these microclimate effects can increase ce-
141 real yields in adjacent strips by 5–15% relative to open-field controls, partially or fully
142 compensating the loss of arable area occupied by the tree strips (Graves et al., 2007).
143 Beyond direct yield effects, AFS deliver broader ecosystem services (like soil organic
144 matter accumulation, carbon sequestration, biodiversity enhancement) that may lead to
145 agri-environment subsidies and carbon credits (Toensmeier, 2017; Faasch and Patenaude,
146 2012).

Nonetheless, economic attractiveness of AFS depends primarily on revenues of AFS biomass. Thereby, revenues are different for the types of biomass and in which value chain it is used. Often AFS biomasses are used energetically, i.e. for heat and power generation, or anaerobic digestion, which typically creates low prices per tonne of biomass.[REF] Alternatively, woody parts of the biomass (i.e., the stem and large branches) can also be used as pulp and paper grade material as well as feedstock for chemical industry or as timber achieving higher prices (Lamers et al., 2011) [EXTEND REFS]. Thus, fostering AFS establishment by maximizing economic returns from AFS biomass requires a strategic approach to product cascading that allocates different biomass fractions to the highest-value applications, while ensuring that the rotation length is aligned with the growth dynamics of the stand and the market demand for different biomass types (Lamers et al., 2011).

Thus, for making educated decisions on AFS establishment and harvesting, the expected biomass quantities at harvest for stem, branches, and residues need to be estimated. Thereby, the shares of the biomass types change with stand age. E.g.~the share of stem weight rises from roughly 55% at age 4 to over 75% at age 10 (Huber et al., 2018; Testa et al., 2014). Therefore, in Section 2.3 growth modelling approaches are reviewed that capture the age-dependent growth dynamics of different biomass types in AFS.

2.3. Biomass growth models

Reliable estimation of stand-level biomass quantities and their age-dependent composition is a necessary prerequisite for any strategic analysis of AFS-based supply chains. Individual-tree biomass is commonly approximated by species-specific allometric power laws relating shoot dry matter to stem base diameter, and these relationships have been calibrated for Central European poplar clones under SRC and SRAFS conditions (Huber et al., 2016, 2018). For supply chain planning, however, the relevant quantity is stand-level biomass per hectare over a multi-decade rotation horizon, which requires a temporal growth model that captures the characteristic sigmoidal accumulation trajectory of woody biomass. Therefore, a set of variables is used summarized in ??.

Table 1: Notation for the biomass growth and harvest yield models.

Symbol	Description
$M^{sl}(t)$	Total above- and belowground dry biomass per ha of AFS (t DM ha ⁻¹) at stand age t
$A(N)$	Stand-level Gompertz asymptote (t DM ha ⁻¹); density- and site-dependent
k	Gompertz growth-rate coefficient (yr ⁻¹)
t_0	Inflection age of stand biomass accumulation (yr)
N	Stand density (trees ha ⁻¹)
C_{site}	Site-quality constant (kg ha ^{β-1}); conservative: 4475, good site: 6471
β	Competition-density exponent; fitted $\hat{\beta} = 0.380$
ϕ_{ag}	Aboveground fraction of total stand biomass; $\phi_{\text{ag}} = 0.89$
$M^{ag}(t)$	Aboveground dry biomass per ha (t DM ha ⁻¹) at age t
$f_p(t)$	asymptotic fraction of AGB attributed to compartment $p \in \{s, b, r\}$ at age t
α_p, δ_p	Intercept and slope of the linear fraction model for compartment p
$q_p(t)$	Merchantable fraction of compartment p biomass at age t (logistic)
r_p	Logistic growth rate for merchantability of compartment p
$t_{50,p}$	Age at which 50% of compartment p biomass is merchantable (yr)
$Y_p(t)$	Merchantable dry biomass of compartment p per ha at harvest age t (t DM ha ⁻¹)
ω	Moisture content at felling (mass fraction); $\omega = 0.55$

175 2.3.1. Stand-level growth model

176 Stand-level aboveground and belowground dry biomass per hectare, $M(t)$, accumu-
177 lates in a right-skewed sigmoidal pattern with a relatively early inflection point, driven
178 by rapid juvenile shoot growth followed by self-thinning and competitive deceleration
179 as canopy closure intensifies (Niemczyk and Thomas, 2021; Testa et al., 2014). This
180 trajectory is described by a Gompertz model:

$$M(t) = A(N) \cdot \exp(-\exp(-k \cdot (t - t_0))), \quad (1)$$

181 where t_0 is the age of maximum instantaneous biomass increment and k determines how
182 quickly the stand approaches its site-specific asymptote $A(N)$. For Central European
183 poplar under medium-rotation AFS management (10–20 yr), analysing empirical data
184 from Huber et al. (2018), Niemczyk and Thomas (2021), and Jha (2018) yields parameter
185 intervals of $k \in [0.17, 0.2]$ and $t_0 \in [9, 10]$ years, depending on local specifications of the
186 AFS under study. Parameter t_0 indicates an upper bound on the optimal rotation age

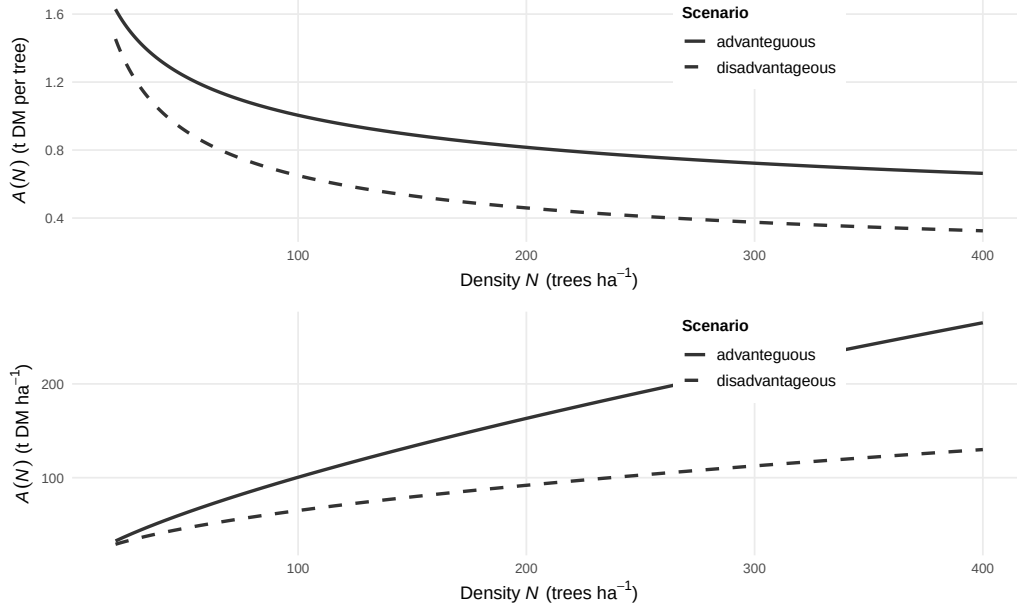


Figure 1: Tree-level and stand-level Gompertz asymptote function $A(N)$ of planting density for the advanteguous ($N = 4$, $\beta = -0.3$) and disadvantageous ($N = 6.5$, $\beta = -0.5$) scenarios.

as growth slows down or diminishes entirely close to this age.

The asymptote $A(N)$ depends largely on planting density and can be modelled by a competition-density power law (Kira et al., 1961):

$$A(N) = C_{\text{site}} \cdot N^{1-\beta}, \quad (2)$$

where C_{site} encodes site quality, N indicates tree density (e.g. trees per hectare), and β is the competition exponent. Equivalently, the per-tree asymptote $A_{\text{tree}}(N) = A(N)/N = C_{\text{site}} \cdot N^{-\beta}$ decreases with density as individual trees have access to less growing space. Parameters C_{site} and β are highly site- and species-specific. For Poplar clones under European conditions, $\beta \in [0.3, 0.5]$ and $C_{\text{site}} \in [4, 6.5]$ (Niemczyk and Thomas, 2021, Kira et al. (1961)). Figure 1 shows asymptote functions according to (2) for advantageous and disadvantageous site conditions.

When assessing potential downstream application of AFS biomass, biomass fraction are to be defined meeting the requirements of downstream consumers. E.g. for timber and biochemical applications, stem diameters of AFS trees need to exceed a minimum threshold in order to qualify as a suitable feedstock. In the following three biomass

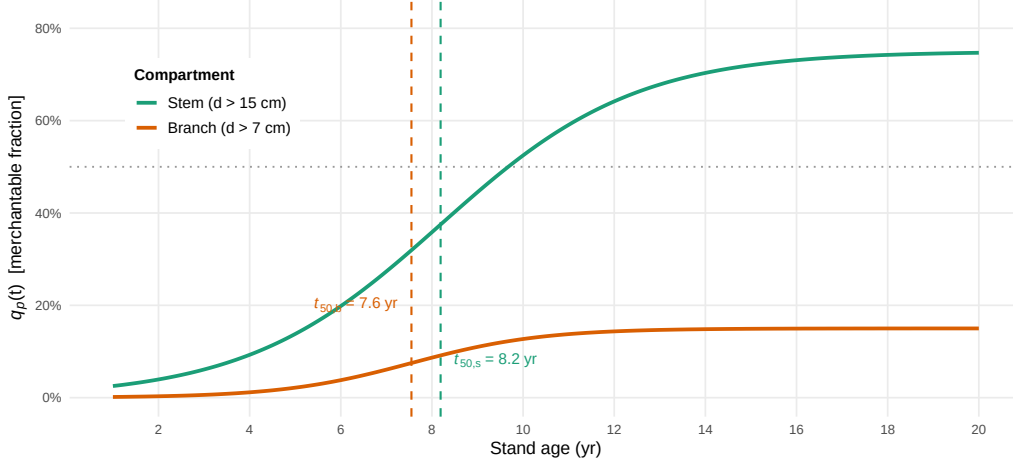


Figure 2: Logistic diameter-class merchantability functions $q_p(t)$ for stem ($d > 15$ cm, green) and branches ($d > 7$ cm, orange); the 50% threshold ages t_{50} are indicated by dashed vertical lines.

fractions are distinguished: stems, branch wood, and residue. The stem fraction consists of stems with a diameter of at least 15 cm, while branch wood requires a diameter of at least 7 cm. The residue fraction comprises are smaller parts of the tree. To subdivide the total AFS biomass modelled by (1) into the three fractions depending on tree age, a logistic growth model is used:

$$q_p(t) = \frac{f_p}{1 + \exp(-r_p \cdot (t - t_{50,p}))}, \quad (3)$$

where $t_{50,p}$ is the age at which 50% of compartment p biomass surpasses the diameter threshold and f_p denotes the asymptotic share of component p for mature trees. Using empirical data from data from Civitarese et al. (2019), Jha (2018) and Niemczyk and Thomas (2021), $r_{stem} = 0.467, \text{yr}^{-1}$, $t_{50,stem} = 8.19$ years, and $f_{stem} = 0.75$ is estimated for the stem fraction (threshold $d > 15$ cm) and $r_{bran} = 0.698 \text{ yr}^{-1}$, $t_{50,bran} = 7.55, \text{yr}$, and $f_{bran} = 0.15$ for the branch fraction (diameter at least 7 cm at insertion). Figure 2 illustrates the development of the fraction shares over stand age.

Combining equations (1)–(3), the dry biomass yield of fraction p per hectare at harvest age t is:

$$\eta_p(t) = q_p(t) \cdot M(t), \quad (4)$$

with all non-merchantable fractions aggregated into the residue fraction. Fresh-weight

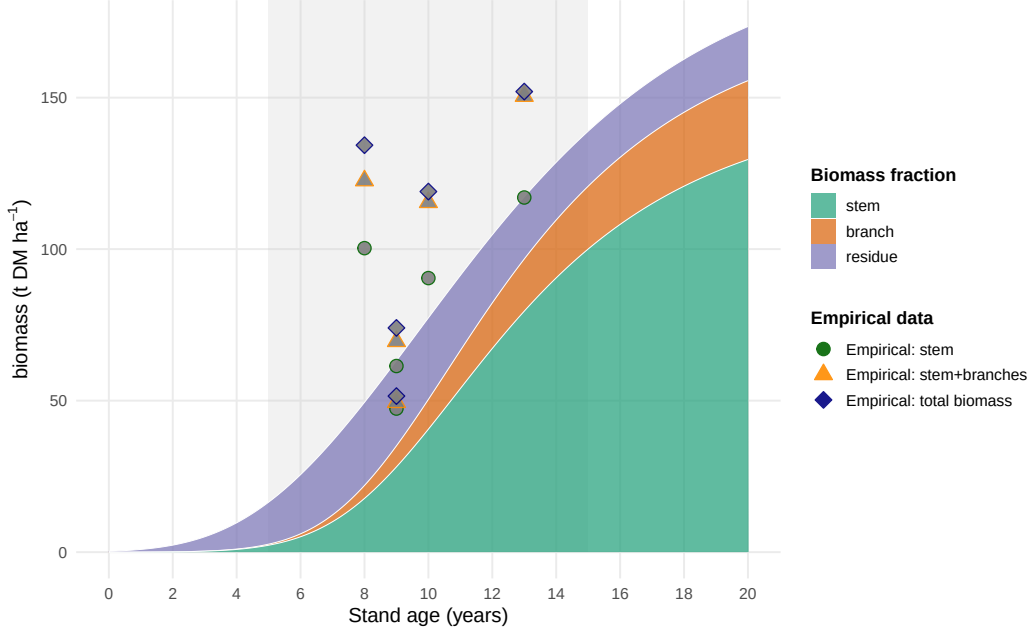


Figure 3: Stand-level aboveground biomass components for poplar under AFS management with $N = 250$ trees ha^{-1} . Empirical AGB observations compiled from Jha (2018), Civitarese et al. (2019), and Niemczyk and Thomas (2021). Grey band: typical medium-rotation harvest window (5–15 years).

yields at felling are obtained from dry matter yields via $\eta_p^{\text{fw}}(t) = \eta_p(t)/(1 - \omega)$, where the moisture content at felling $\omega = 0.55$ is taken from Civitarese et al. (2019). Figure 3 shows the fractions' stacked biomass quantities $\eta_p(t)$ depending on stand age compared with empirical observations from Jha (2018)

2.4. Biomass supply chain design

The design and planning of biomass supply chains has attracted substantial research attention over the past two decades, driven by the need to ensure cost-efficient and sustainable feedstock provision for bioenergy and biorefinery applications. Several systematic reviews provide comprehensive overviews of this literature. Cambero and Sowlati (2014) survey economic, social, and environmental assessment and optimisation studies for forest biomass supply chains, identifying MILP as the dominant modelling approach and noting a growing trend towards multi-objective formulations integrating life cycle assessment.

Yue et al. (2014) review biomass-to-bioenergy and biofuel supply chain models with a focus on multi-scale modelling, uncertainty treatment, and sustainability metrics, and conclude that the coupling of upstream feedstock dynamics with downstream network design remains an underexplored area. Malladi and Sowlati (2018) survey logistics-specific features of biomass supply chain models—seasonal availability, moisture content degradation, preprocessing steps—and highlight that most models treat biomass as a homogeneous commodity, neglecting quality differentiation. Zahraee et al. (2020) provide a recent meta-review of 140+ studies, confirming that cost minimisation is still the dominant objective, and that multi-product cascading formulations are rare.

The structure of biomass supply chain models typically encompasses three decision layers: (i) strategic network design (facility location, capacity sizing, technology selection), (ii) tactical planning (harvest scheduling, seasonal flows, inventory), and (iii) operational routing. The following paragraphs discuss key individual contributions and their specific methodological advances.

Ekşioğlu et al. (2009) develop one of the earliest large-scale MILP models for biomass-to-biorefinery supply chain design, integrating facility location, capacity sizing, and multimodal transport for lignocellulosic feedstocks. Their model minimises total system cost and introduces binary facility-opening variables that have since become standard in the field. A key limitation, however, is that biomass supply is treated as an exogenous input, so feedstock availability is not endogenised as a function of land-use or growth decisions. Mansoornejad et al. (2013) extend the strategic design perspective to forest biorefineries by integrating product portfolio design with supply chain configuration in a single MILP. Their model explicitly accounts for product conversion efficiencies along different processing pathways, enabling joint optimisation of process technology selection and logistics.

De Meyer et al. (2014) introduce the OPTIMASS framework, a generic MILP for biomass-based supply chain optimisation that explicitly models product quality cascades: higher-quality biomass classes can substitute for lower-quality demands downstream, whereas the reverse is infeasible. This cascade hierarchy, formalised through ordered product sets, is particularly relevant for woody biomass where stem, branch, and residue fractions command different market prices. The companion t-OPTIMASS

extension (De Meyer et al., 2016) incorporates temporal dynamics by explicitly representing biomass growth and regeneration over multiple periods, enabling simultaneous optimisation of harvest timing and network flows. However, in both OPTIMASS formulations the growth dynamics are represented as exogenous time-series rather than as age-structured stand models, so rotation-length decisions and age-dependent biomass quality are not endogenised. Arabi et al. (2019) demonstrate multi-period MILP extensions for bio-based supply chains with production dynamics, but focus on microalgae rather than woody biomass and do not address product quality cascades.

Sharma et al. (2013) identify supply-side uncertainty (yield variability, seasonal availability, moisture content) as the most critical source of risk in biomass supply chains. In response, a strand of literature has developed stochastic and robust optimisation extensions. Zahraee et al. (2020) document that two-stage stochastic MILP and robust optimisation approaches now account for roughly 30% of the modelling literature, with biomass yield uncertainty and price volatility as the dominant stochastic parameters.

Biomass supply chain research focusing specifically on agroforestry or SRC contexts is comparatively sparse. Espinoza et al. (2017) note that most supply chain models assume continuous feedstock availability and do not represent the periodic harvest cycles characteristic of coppice systems. Lamers et al. (2011) analyse feedstock design principles for lignocellulosic supply chains and argue that the allocation of different biomass fractions to valorisation pathways—product cascading—is a key lever for economic viability, yet is rarely optimised jointly with network design decisions. The model developed in Section 3 of this paper directly addresses this gap by endogenising stand age dynamics through binary age-lag variables and coupling component-specific yield coefficients with a product quality cascade.

Table 2 summarises the key modelling features of the reviewed contributions.

3. MILP Model for Agroforestry Supply Chain Design

Building on the biomass growth models introduced in Section 2.3 and the supply chain design literature reviewed in Section 2.4, this section describes the AFS supply chain design problem (AFS-SCD), a mixed-integer linear programming (MILP) model for the integrated design and operation of an agroforestry biomass supply chain with

Table 2: Key modelling features of selected biomass supply chain design studies.

Study	Objective	Multi-product	Quality cascade	Age dynamics	Uncertainty	Level
Eksioglu et al. (2009)	Cost min.	—	—	—	—	Strategic
Sharma & Taaffe (2013)	Cost min.	Partial	—	—	—	Strategic/Tactical
Mansoornejad et al. (2013)	Profit max.	✓	—	—	—	Strategic
Cambero & Sowlati (2014)	Cost/GHG/jobs	—	—	—	Stochastic	Strategic
Yue et al. (2014)	Cost/GHG	✓	—	—	—	Multi-scale
De Meyer et al. (2015)	Cost min.	✓	✓	Exogenous	—	Strategic/Tactical
De Meyer et al. (2016)	Cost min.	✓	✓	Exogenous TS	—	Multi-period
Lamers et al. (2015)	Cost/quality	✓	Partial	—	Scenario	Strategic
Espinoza et al. (2017)	Sustainability	✓	—	—	Scenario	Strategic
Malladi & Sowlati (2018)	Cost min.	Partial	—	—	—	Tactical/Oper.
Arabi et al. (2019)	Profit max.	✓	—	Exogenous	—	Multi-period
Zahraee et al. (2020)	Multi-obj.	✓	—	—	Stochastic	Strategic/Tactical
This paper	NPV max.	✓	✓	Endogenous	Scenario	Strategic/Multi-period

multiple AFS biomass types (i.e., stem, branch, and residue fractions) used as feedstocks for potential consumers in chemical, pulp and paper as well as energy industry.

3.1. Sets, Indices, and Time Structure

The model considers a set of agroforestry sites $\mathcal{I}$, storage/pre-treatment facilities $\mathcal{J}$, consumer sites $\mathcal{K}$, and biomass types $\mathcal{P} = \{stem, branches, residues\}$, whose growth models are outlined above. Time is discretised into annual periods $t \in \mathcal{T} = \{1, \dots, T_{\max}\}$. To reduce complexity, reflect legal definitions of SRAFS and LRAFS as well as discard implausible options (e.g. extremely long rotation cycles), $A^{\min}$ and $A^{\max}$ define minimum and maximum stand ages. Thus, harvest are possible in periods $\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T_{\max}\}$. To consistently model establishment and consecutive harvest decisions, the following arc sets are defined: $\mathcal{S}^{\text{est}}$ defines potential establishments of AFS. If an AFS is established in period $t \in \mathcal{T}$, arc $(0, t)$ is selected. Harvest set $\mathcal{S}^{\text{harv}}$ summarizes all potential consecutive harvest periods within the age interval $[A^{\min}, A^{\max}]$. I.e., if an AFS is established

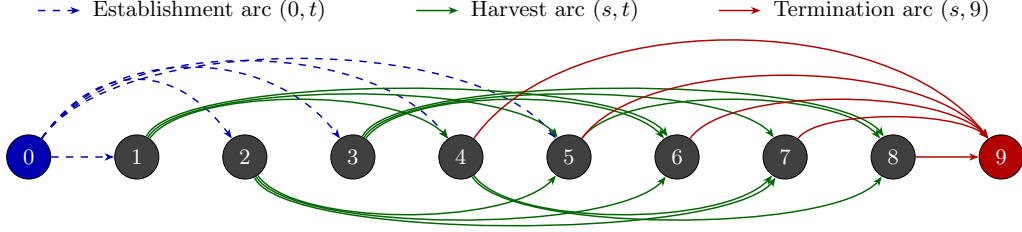


Figure 4: Network structure of set $\mathcal{S}$ for $T = 8$, $A_{\min} = 3$, $A_{\max} = 5$. Dashed blue arcs: establishment arcs $(0, t)$, $t \in \{1, \dots, T - 1\}$. Solid green arcs above: harvest arcs (s, t) with odd s ; below: harvest arcs with even s . Solid red arcs: termination arcs $(s, T + 1)$, $s \in \{A_{\min} + 1, \dots, T\}$.

or harvested in period s and next harvest is in period t , arc (s, t) is selected. Finally, if there is no subsequent harvest after period t , the AFS is terminated by selecting arc $(t, T^{\max} + 1)$. Termination arcs are summarized in set $\mathcal{S}^{term}$. All arc sets are formally defined as follows:

$$\begin{aligned}\mathcal{S}^{est} &= \{(0, t) \mid t \in \{1, \dots, T^{\max} - A^{\min}\}\}, \\ \mathcal{S}^{harv} &= \{(s, t) \mid s, t \in \mathcal{T} \wedge t - s \in \{A^{\min}, \dots, A^{\max}\}\}, \\ \mathcal{S}^{term} &= \{(s, T^{\max} + 1) \mid s \in \{A^{\min} + 1, \dots, T_{\max}\}\}, \\ \mathcal{S} &= \mathcal{S}^{est} \cup \mathcal{S}^{harv} \cup \mathcal{S}^{term}.\end{aligned}$$

Figure 4 illustrates this network structure of all arc sets for $T = 8$, $A^{\min} = 3$, $A^{\max} = 5$.

3.2. Decision Variables

Temporal flow Variables z_{ist} indicate on which quantity of the area of site i an AFS is operated between $(s, t) \in \mathcal{S}$. Depending on the arc type, this implies establishment, harvest, or termination. Selected arcs build a network flow over the complete planning horizon from $t = 0$ to $T^{\max} + 1$. Figure 5 illustrates a simple network flow with establishment of an AFS in period 1 and harvests in periods 5 and 8.

Spatial flow variables X_{ijpt} and $X_{jkpp't}$ represent transport of products from AFS sites to storage/pre-treatment facilities and from there to consumers, respectively, while S_{jpt} captures end-of-period inventory at facility j . Table 3 summarises all sets, decision variables, and parameters used in the MILP formulation, while η_{pa} refers to (4).

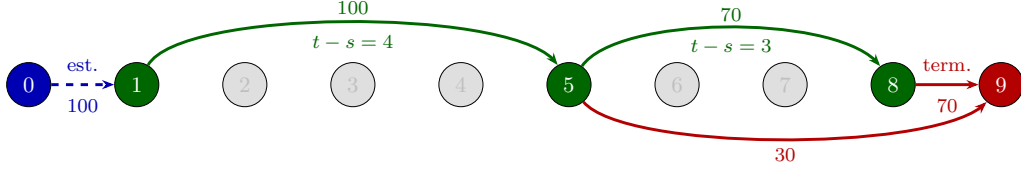


Figure 5: Example of a feasible path through the network for $T = 8, A_{\min} = 3, A_{\max} = 5$: establishment in period 1, harvests in periods 5 and 8, termination at $T + 1$.

3.3. Objective Function

As discussed in Section 2.1, the economic attractiveness of AFS supply chains is modelled from a global perspective taking into account all processes, activities, and participants involved in finally serving customer markets with AFS-based feedstocks. Therefore, revenues for the different biomass fractions and are weighed against costs occurring at all value-adding stages of the supply chains. Revenues created by selling AFS-based feedstocks to different industrial consumers are counterweighted by establishment, opportunity, harvest, transport, and storage costs of landowners, agricultural firms as well as biomass logistics companies. The model therefore maximises total profit of all participants in the AFS supply chain over the planning horizon as follows:

$$\begin{aligned}
 \max Z = & \sum_{k \in \mathcal{K}} \sum_{p \in \mathcal{P}} R_{kp} \cdot \sum_{j \in \mathcal{J}} \sum_{(p', p) \in \mathcal{Q}} \sum_{t \in \mathcal{T}} X_{jkp'pt} \\
 & - \sum_{i \in \mathcal{I}} c_i^{est} \cdot \sum_{(0, t) \in \mathcal{S}^{est}} z_{i0t} \\
 & - \sum_{i \in \mathcal{I}} (c_i^{main} + c_i^{opp}) \cdot (t - s) \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}} c_i^{harv} \cdot \sum_{(s, t) \in \mathcal{S}^{harv}} z_{ist} \\
 & - \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_p^{tr-raw} \cdot d_{ij} \cdot X_{ijpt} \\
 & - \sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K}} \sum_{(p, p') \in \mathcal{Q}} \sum_{t \in \mathcal{T}} c_{p'}^{tr-pre} \cdot d_{jk} \cdot X_{jkpp't} \\
 & - \sum_{j \in \mathcal{J}} \sum_{p \in \mathcal{P}} \sum_{t \in \mathcal{T}} c_j^{stor} \cdot S_{jpt}
 \end{aligned} \tag{5}$$

Here, R_{kp} are product- and consumer-specific revenues for each ton of biomass fraction (€/t). The opportunity cost term penalises the land use when AFS are established.

Table 3: Notation for the MILP model.

Sets & Indices	
$i \in \mathcal{I}$	Agroforestry sites
$j \in \mathcal{J}$	Storage / pre-treatment facilities
$k \in \mathcal{K}$	Consumer sites
$p \in \mathcal{P}$	Product types (stem, branches, residues)
$(p, p') \in \mathcal{Q}$	Admissible product-substitution pairs (quality cascade)
$\mathcal{T} = \{1, \dots, T\}$	Planning periods (years)
$\mathcal{T}^{\text{harv}} = \{A^{\min} + 1, \dots, T\}$	Feasible harvest periods
$\mathcal{S}^{\text{est}} = \{(0, t) \mid t \in \{1, \dots, T - A^{\min}\}\}$	Establishment arcs: dummy source to period t
$\mathcal{S}^{\text{harv}} = \{(s, t) \mid s, t \in \mathcal{T}, t - s \in \{A^{\min}, \dots, A^{\max}\}\}$	Harvest cycles within age-lag bounds
$\mathcal{S}^{\text{term}} = \{(s, T + 1) \mid s \in \{A^{\min} + 1, \dots, T\}\}$	Termination arcs to dummy sink $T + 1$
$\mathcal{S} = \mathcal{S}^{\text{est}} \cup \mathcal{S}^{\text{harv}} \cup \mathcal{S}^{\text{term}}$	Age-lag arc set (establishment $\cup$ harvest $\cup$ termination)
Decision Variables	
$z_{ist} \geq 0$	Area of site i with AFS operation between s and t
$X_{ijpt} \geq 0$	Transport flow (t) of p from site i to facility j in t
$X_{jkpp't} \geq 0$	Flow of p from facility j to consumer k satisfying demand p' in t
$S_{jpt} \geq 0$	Inventory (t) of product p at facility j end of period t
Parameters	
T	Length of planning horizon (years)
$A^{\min}, A^{\max}$	Minimum / maximum stand age at harvest (years)
AREA_i	Area of site i (ha)
η_{pa}	Yield coefficient (t/ha) of fraction p at stand age a , derived from (4)
R_{kp}	Revenue (€/t) for product p at consumer k
C_i^{est}	Establishment cost (€/ha) at site i
C_i^{opp}	Annual opportunity cost (€/ha) at site i
C_i^{harv}	Harvest & refitting cost (€/ha) at site i
$c^{\text{tr-raw}}$	Transport cost rate (€/t km) for raw biomass
$c^{\text{tr-pre}}$	Transport cost rate (€/t km) for pre-treated biomass
c_j^{stor}	Storage holding cost (€/t) at facility j
d_{ij}, d_{jk}	Distance (km) between nodes
$\text{CAP}_j^{\text{stor}} / \text{CAP}_j^{\text{proc}}$	Storage and processing capacity (t) at facility j
$D_{kpt}^{\max}$	Maximum demand (t) of fraction p at consumer k in period t

331 Thereby, c_i^{opp} can be negative or positive depending on the site-specific conditions. It
332 comprises effect of lost field crop revenues due to AFS as well as higher yields of remain-

ing field crop areas e.g. due to wind and water erosion protection by AFS. While the
 assessment of lost field crop revenues is usually well predictable, the economic evaluation
 of positive ecological effects of AFS (e.g. on neighboring crop yields) is more difficult.
 Thus, c_i^{opp} can also contain regulatory subsidies to support establishing and maintaining
 AFS. Transport cost terms mirror the two-stage flow structure of the AFS supply chain
 with differentiated rates for raw and pre-treated biomass depending on the biomass type.
 Cost rates differ because as e.g. the density and water content of the shipped biomasses
 differ or because specialized equipment (e.g. for log transports) is used [REFERENCES].
 Moreover, site-specific establishment cost for planting the AFS trees is considered, next
 to yearly maintenance and harvesting costs. All these costs scale with the size of a
 site ($AREA_i$) measured in hectar. Maintenance cost comprise e.g. costs for trimming,
 replanting, and pruning unnecessary shoots. Storage costs are considered if biomasses
 are stored for more than a year. Storage cost rate c_j^{stor} measures the costs for storing
 a ton of biomass for one year and comprises direct storage cost (e.g. for storage site
 maintenance) as well as opportunity cost due to potential material losses and capital
 commitment. Biomasses harvested and consumed in the same year incur no storage cost,
 as usually AFS are harvested at the beginning of a year. Harvested biomass has to dry
 during spring and summer such that the biomasses can be used in autumn or winter
 the same year. As under-year drying is necessary for most industrial applications and
 beneficial for transport economics, drying storage in the year of harvest is not associated
 with additional cost.

3.4. Constraints

Constraints (6) limits the area dedicated to AFS operations to the maximum size
 of site i . Constraints (7) enforce flow conservation in each period. If AFS systems of
 size z_{ist} are established or harvested in period $t \in \mathcal{T}$, there needs to be a succeeding
 activity (harvest or termination) of this area. This way, (7) guarantee valid (possibly
 empty) AFS activity over the complete time horizon for each site. Constraints (8) links
 the harvest-cycle variables z_{ist} to physical yield quantities $\eta_{p(t-s)}^{fw}$ for age $t - s$, which
 are pre-computed using (1)–(4) scaled by moisture content ω . Thus, we assume constant
 yields per hectar for each site i . Note that this simplification can be dropped and site-
 specific yields $\eta_{ip(t-s)}^{fw}$ could be introduced without increasing numerical complexity of the

364 AFS-SCD . Constraints (9) link harvest quantities to transport flows and ensure inven-
 365 tory balance at each facility. Thereby, harvest quantities can be smaller than available
 366 AFS biomasses allowing partial harvesting. Constraints (10)–(11) enforce storage and
 367 processing capacity limits at each facility j . Constraints (12) implement product usage at
 368 multiple consumption sites k . Thereby, $(p', p) \in \mathcal{Q}$ defines a product mapping allowing
 369 high-quality product p' (e.g. stems) satisfying demand of a lower-quality products p up
 370 to maximum demand $D_{kpt}^{\max}$.

$$\sum_{(0,t) \in \mathcal{S}^{est}} z_{i0t} \leq AREA_i \quad \forall i \in \mathcal{I} \quad (6)$$

$$\sum_{(s,t) \in \mathcal{S}} z_{ist} = \sum_{(t,u) \in \mathcal{S}} z_{itu} \quad \forall i \in \mathcal{I}, t \in \mathcal{T} \quad (7)$$

$$\sum_{j \in \mathcal{J}} X_{ijpt} \leq \sum_{(s,t) \in \mathcal{S}^{harv}} \eta_{p(t-s)}^{fw} \cdot z_{ist} \quad \forall i \in \mathcal{I}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (8)$$

$$S_{jpt} = S_{jp,t-1} + \sum_{i \in \mathcal{I}} X_{ijpt} - \sum_{k \in \mathcal{K}, (p,p') \in \mathcal{Q}} X_{jkpp't} \quad \forall j \in \mathcal{J}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (9)$$

$$\sum_{p \in \mathcal{P}} S_{jpt} \leq CAP_j^{stor} \quad \forall j \in \mathcal{J}, t \in \mathcal{T} \quad (10)$$

$$\sum_{i \in \mathcal{I}, p \in \mathcal{P}} X_{ijpt} \leq CAP_j^{proc} \quad \forall j \in \mathcal{J}, t \in \mathcal{T}^{harv} \quad (11)$$

$$\sum_{j \in \mathcal{J}, (p', p) \in \mathcal{Q}} X_{jkp'pt} \leq D_{kpt}^{\max} \quad \forall k \in \mathcal{K}, p \in \mathcal{P}, t \in \mathcal{T}^{harv} \quad (12)$$

$$z_{ist}, X_{ijpt}, X_{jkp'pt}, S_{jpt} \geq 0 \quad \forall i, j, k, p, p', t \quad (13)$$

371 In summary, AFS-SCD constitutes are large-scaled LP which allows to decide on
 372 which fraction of a site, AFS are established and how the AFS operations are scheduled
 373 in a multi-year planning horizon. The AFS-SCD takes a supply chain perspective and
 374 maximizes total profit of all actors involved in these processes. Thus, it does not explain
 375 how the profits are split e.g. between farmers, AFS service providers and logistics service
 376 providers. The continuous modelling of AFS areas in z_{ist} leads to a comparatively easy
 377 to solve problem. On the other side, it is unclear which part of a site's total area is
 378 chosen for AFS operations if $z_{ist} < AREA_i$, which holds for establishment as well as
 379 harvesting and termination.

380 4. Case Study

381 The AFS-SCD is intended to support the design of regional agroforestry biomass sup-
 382 ply chains fostering regional bioeconomy clusters thereby connecting multiple industrial
 383 sectors. The proposed AFS-SCD model is instantiated on a real-world biomass supply
 384 chain case study located in the tri-state region of Saxony-Anhalt, northern Thuringia,
 385 and northern Saxony, Germany. This region is characterized by a high density of agri-
 386 cultural land usage as well as cluster of paper and wood as well as chemical industry.
 387 The regional chemical industry relies on fossil resources so far, but is under substantial
 388 transformative pressure which led to investments in sustainable production processes
 389 requiring sustainable regional bio-based feedstocks. Figure 6 shows agricultural produc-
 390 tion sites along with the set of potential wood-processing hubs, and the regional users of
 391 AFS biomasses from chemical, paper and wood processing industry as well as the energy
 392 sector.

393 The supply chain comprises 2239 agricultural production parcels which are grouped
 394 into $|\mathcal{I}| = 145$ potential AFS production sites. Clustering is based on geographical dis-
 395 tances between parcel centroid using a k-means approach in order to reduce problem
 396 size. For distance calculations, for each site a centroid is calculated. Pre-treatment and
 397 intermediate storage facilities $\mathcal{J}$ are located on existing wood processing facilities as well
 398 as potential new log or wood chip yards. All harvested biomass quantities at AFS sites
 399 are transported to logistics hubs first. At the hubs sorting and transshipment processes
 400 take place before biomass types are forwarded to the consumer facilities. Storing and
 401 transshipment capacities at the hubs are assumed to be constant in time and universally
 402 set to 4.0909×10^4 kt per year. Finally, $|\mathcal{K}| = 28$ consumer sites are included. Each con-
 403 sumer site belongs to an existing facility from chemical, paper/wood processing industry,
 404 or the energy sector. Distances between the facilities d_{ij} and d_{jk} are calculated as road
 405 distances between the facilities' coordinates. For each consumer facility, an yearly upper
 406 demand capacity for each of the three distinguished biomass product types are given
 407 along with consumer-specific prices. The following biomass types are distinguished:

- 408 • **P1: stem wood – chemical/paper grade:** long-fibre wood chips from stems
 409 suitable for conversion in a local biorefinery and paper factory. In total, a yearly
 410 demand of 620 kt and an average prices of 45 € is assumed.

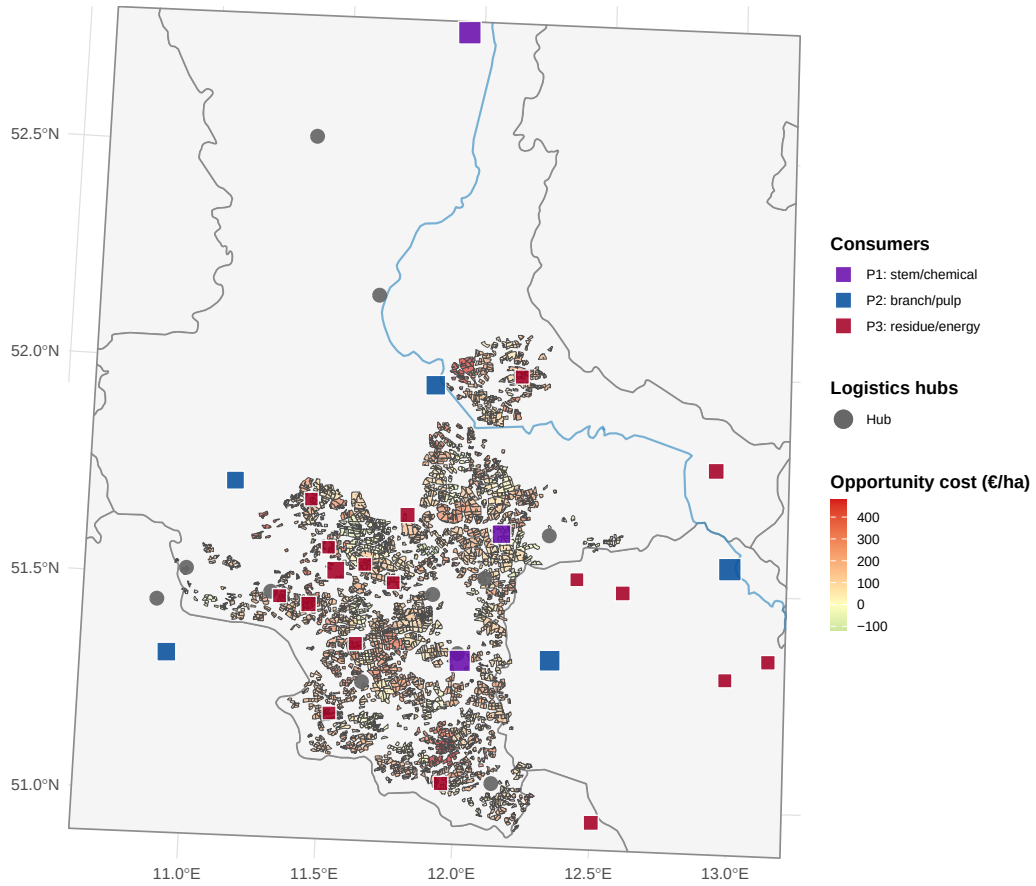


Figure 6: AFS supply chain network in Saxony-Anhalt. Grey lines: German federal state boundaries. Blue lines: major rivers. Coloured polygons: potential AFS parcels grouped by site (colours encode cluster membership; no legend). Orange symbols: pre-treatment facilities (circle = sawmill, triangle = logistics hub, size proportional to processing capacity). Coloured squares: consumer sites (violet: P1 chemical/pulp, blue: P2 pulp/paper, red: P3 energy/biogas, size proportional to total demand).

- 411 • **P2: branch wood – sawmill/pulp grade:** medium/short-fibre wood chips
412 from larger branches suitable for mechanical processing in sawmills and lower-grade
413 pulping. Demand is distributed across 6 smaller facilities and accumulates to 410
414 kt/year. Gate prices range from 28-38 €/t.
- 415 • **P3: residue – energy grade:** residual wood fractions (chips, bark, off-cuts) used
416 for heat and electricity generation or lower-grade pellet production. Demand is
417 accumulates among 24 consumers to 132 kt/year. Gate prices range from 18-24
418 €/t.

419 Table 7 in the appendix summarizes the consumer-specific demand data. It is assumed
420 that the biomass product types show a quality cascade which means that the demand of
421 pulp grade (P2) can also be satisfied by chemical/paper grade biomass (P1). Likewise,
422 energy grade demand (P3) can be satisfied by all biomass types. Note that constant
423 demands over the planning horizon are assumed, although an increasing biomass demand
424 can be expected due to the green transition in those industries. From this perspective, a
425 rather conservative market demand scenario is assumed.

426 Growth model introduced in subsection 2.3 is used to model biomass quantities de-
427 pending on age and the area dedicated to AFS. The growth model is parameterized for
428 poplar clones at central European conditions as introduced above with a planting density
429 of $N = 400$ trees per hectar. If AFS are established, they have to be harvested within
430 $A^{min} = 3$ to $A^{max} = 20$ years allowing a wide range of potential harvesting strategies
431 between short-rotation and long-rotation AFS. As the total planning horizon is $|\mathcal{T}| = 40$
432 years multiple harvesting cycles can be established. The sites do not differ with respect
433 to growth conditions as the climatic and soil quality variation within the comparatively
434 small region is rather limited. However, the sites differ with respect to size (given in
435 hectar) and opportunity cost rates (given in € per hectar and year). Site-specific oppor-
436 tunity cost rates c_i^{opp} is normally distributed with mean 521 and standard deviation 391.
437 For most sites, substantial opportunity costs are assumed reflecting the loss in field-crop
438 revenue, risk premiums of landowners as well as higher costs for field crop maintenance.
439 The 145 sites with negative opportunity cost rates reflect sites where AFS lead to an
440 increase in field crop productivity on the remaining field area. The variation in opportu-
441 nity cost rates to study the effect of opportunity cost on site selection for AFS and gives

valuable insights particularly for communication with farmers and politics. Subsidies for establishing AFS are one of the core instruments to foster AFS. The height and effectiveness of those subsidies, however, is controversially disputed. Remaining cost parameters are universally set for all sites and comprise maintenance cost rates c_i^{main} (€ per year and hectare), establishment cost c_i^{est} (€ per hectare) and harvesting cost c_i^{harv} (€ per hectare). Likewise, transport cost rates for harvested biomasses c_p^{tr-raw} and pre-treated/sorted biomasses $c_{p'}^{tr-raw}$ are set universally. All scalar parameters are shown in Table 4.

Table 4: Parameter values for the AFS-SCD case study. Transport costs refer to road distance

Parameter	Description	Base-case value	Unit
Index sets			
$ \mathcal{T} $	Planning periods	40	yr
$ \mathcal{I} $	Sites	145	–
$ \mathcal{J} $	Pre-treatment / storage nodes	11	–
$ \mathcal{K} $	Consumers	28	–
$ \mathcal{P} $	Product grades	3	–
Biomass dynamics			
$A^{\min}$	Minimum harvest age (years)	3	yr
$A^{\max}$	Maximum stand age (years)	20	yr
Cost parameters			
c^{tr-raw}	Raw transport cost	0.08	€ t ⁻¹ km ⁻¹
c^{tr-pre}	Pre-processed transport cost	0.06	€ t ⁻¹ km ⁻¹
c_i^{est}	Establishment cost rate	2500	€ ha ⁻¹ yr ⁻¹
c_i^{harv}	harvest cost rate	150	€ ha ⁻¹ yr ⁻¹
c_i^{main}	Maintenance cost rate	10	€ ha ⁻¹ yr ⁻¹
c_i^{opp}	aver. opportunity cost	521	€ ha ⁻¹ yr ⁻¹
Revenue parameters			
R_{P1}	Mean gate price P1	45	€ t ⁻¹
R_{P2}	Mean gate price P2	33	€ t ⁻¹
R_{P3}	Mean gate price P3	21	€ t ⁻¹

4.1. Optimization results

The AFS-SCD model is solved for the instance data described above with Gurobi. The model selects a subset of sites for AFS establishment and identifies the optimal harvest schedules, intermediate facility utilisation, and product allocation across the planning horizon.

Figure 7 shows the spatial configuration of the profit-optimal supply chain.

In total, on 38 sites AFS are established which corresponds to about 26 % of all sites and 24 % of the total farm area. Over the planning horizon, 2.6368×10^4 kt of AFS biomasses are produced on these sites which suffices to satisfy 57 % of total demand. ?? shows the harvested biomasses over time along with total demand of all consumer sites.

During the first 10 years no AFS biomasses are harvested as trees of established AFS in first years grow up to a certain size before first harvest. Thus, about 25% of consumers' total biomass demand cannot be satisfied due to this grow-up period. Afterwards, a cyclical pattern appears with a harvesting peak around year 20 and a breakdown in year 28 before plateau is reached from year 35-40. Thereby, the harvesting quantities of stems and branches show a similar pattern while quantities of harvested energy grade biomass are rather stable and close to the demand limit over the planning horizon. Thus, rather long-rotation AFS are established as ?? confirms showing the histogram of harvesting ages across all sites.

The histogram illustrates that majority of harvest events occur at stand ages between 14–18 years, indicating that long-rotation AFS seem to be more profitable in this setting. Short-rotation cycles (<10 years) occur only in marginal frequencies. This optimal rotation length might be caused either due to the limited demand of energy-grade biomass or the price gaps between biomass types or a combination of effects. The quantity of energy-grade biomass created in long-rotation AFS suffices to cover total energy-grade biomass demand almost completely. Rotation length is focused on production of chemical-grade biomass where highest prices are assumed. Focusing on chemical industry supply requires a production of trees with a comparatively large stems. This strategy maximizes revenue, but in the same time increases opportunity costs as wood harvests appear in larger time intervals. ?? presents the aggregate cost–profit waterfall over all periods and actors.

Total revenue is dominated by P1 deliveries to the two large pulp mills. Opportu-

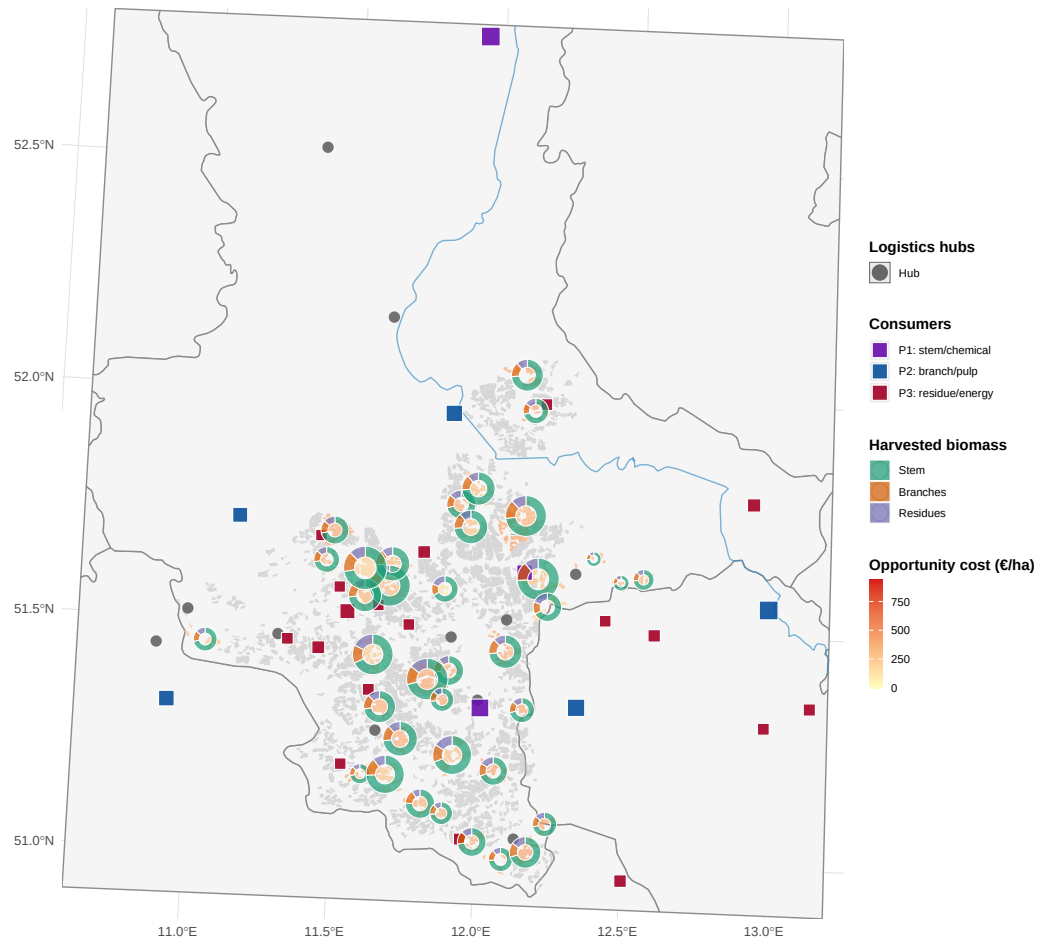


Figure 7: Site selection and biomass production in optimal supply chain configuration. Green polygons: active sites with biomass harvest; grey polygons: non-selected sites. Pie charts at active sites show the product mix of harvested biomass (stem = P1, branches = P2, residues = P3), scaled by total harvested volume.

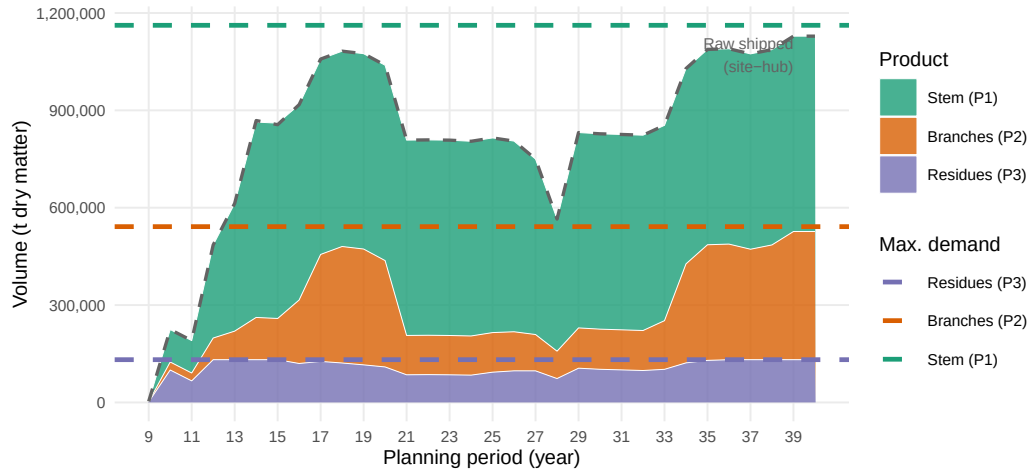


Figure 8: Total biomass delivered to consumers by product type over the planning horizon. Stacked areas show period-level volumes of stem (P1), branches (P2), and residues (P3) flowing from storage hubs to all consumers. Dashed line: total harvested raw biomass shipped from sites to hubs.

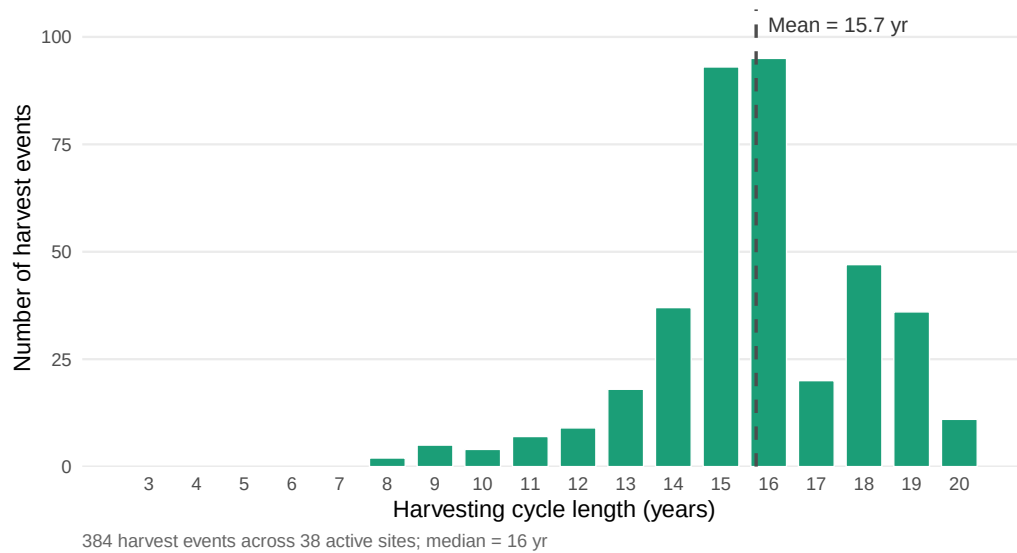


Figure 9: Distribution of harvesting cycle lengths at active agroforestry sites. Each bar represents the number of individual harvest events (over all periods and active sites) whose cycle length (stand age at harvest, i.e. $t - s$) falls in that year bin.

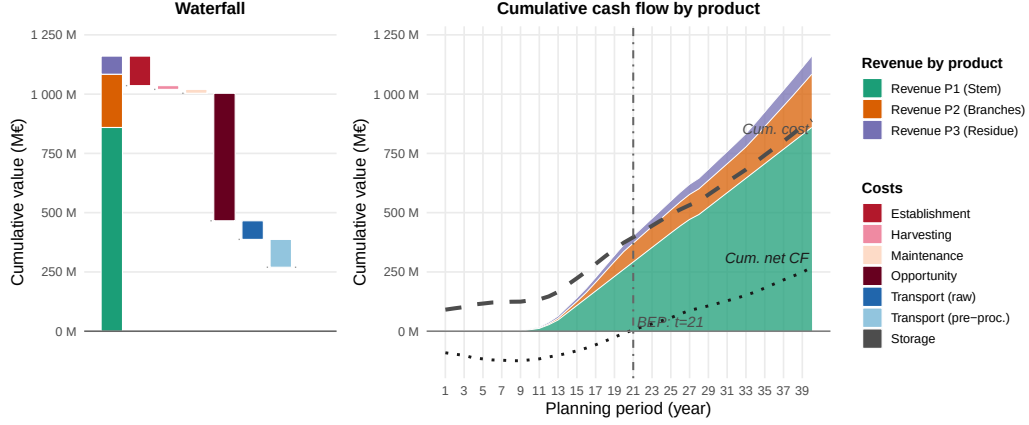


Figure 10: Total cost–profit breakdown across all actors and years. The left panel (waterfall) decomposes the objective value into revenue and individual cost categories; the right panel shows the period-level trajectory of revenues and stacked costs.

480 nity and establishment costs are the main cost drivers whereby opportunity cost clearly
 481 dominate.

482 The site-level revenue–cost decomposition in Figure ?? confirms that profitability is
 483 highly heterogeneous: a small number of large sites with low opportunity costs and short
 484 distances to P1 consumers account for the bulk of total net profit, while several marginal
 485 sites operate near break-even. Figure ?? reveals a clear negative relationship between
 486 opportunity cost rate and site profitability, with sites exceeding $300 \text{ € ha}^{-1} \text{ yr}^{-1}$ rarely
 487 selected in the optimal solution.

488 5. Sensitivity analysis

489 To assess the robustness of the baseline results and to identify the principal drivers
 490 of supply chain profitability, a factorial sensitivity analysis is conducted. The analy-
 491 sis follows a full-factorial design over three scenario dimensions: opportunity cost level
 492 ($\text{€ ha}^{-1} \text{ yr}^{-1}$), a cost multiplier applied to all variable cost parameters (establishment,
 493 harvest, transport, and storage), and a revenue multiplier applied to all gate prices. The
 494 resulting $2 \times 2 \times 2 = 8$ scenarios are solved independently in addition to the baseline,
 495 yielding nine comparable instances (Table ??).

496 Table ?? summarises the key performance indicators across all scenarios. Total profit



Figure 11: Total revenue and cost breakdown per active agroforestry site over the full planning horizon. Each bar represents one active site; revenue (green, above zero axis) and cost components (stacked below) are expressed in thousand euros per hectare for comparability across sites of different size. Sites are ordered by descending net margin.

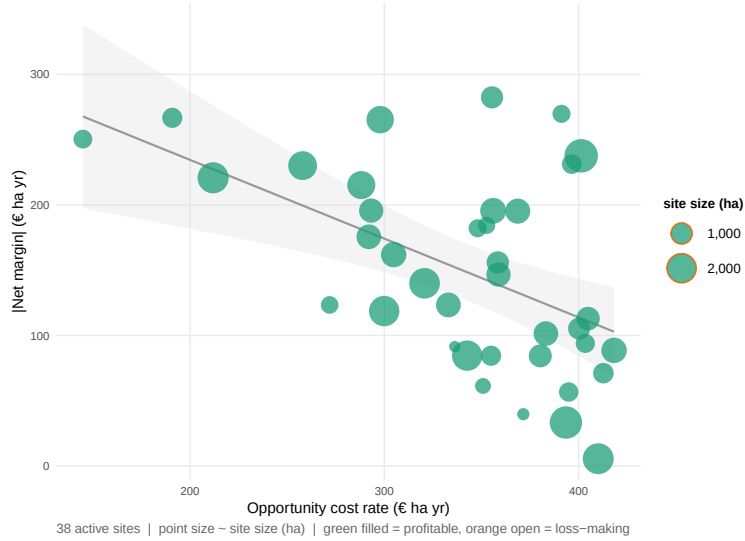


Figure 12: Scatterplot of active agroforestry sites: opportunity cost rate (€ ha yr) on the x-axis versus demand-weighted average delivery distance (km) on the y-axis. Point size is proportional to the site's net margin (€ ha yr); sites with negative net margin are shown as open circles. A LOESS smoothing line indicates the joint trend.

per hectare per year is most sensitive to the revenue multiplier: elevated gate prices (revenue multiplier = 1.5) substantially improve profitability, while reduced prices (multiplier = 0.5) push marginal sites into loss territory. The supply chain remains profitable in all scenarios with full or elevated revenue even when opportunity costs are high. Scenarios combining reduced revenue (multiplier = 0.5) with high opportunity costs consistently yield negative net profits, indicating that the economic viability of AFS-based supply chains is critically conditional on achieving sufficient product-quality premiums from P1 offtakers. The number of active sites and total wood throughput respond monotonically to the revenue multiplier, confirming that higher gate prices incentivise the activation of marginal sites.

The OLS regression reported in Figure ?? and Table ?? decomposes profit variation across scenarios and sites into effects of four explanatory variables: opportunity cost rate, cost level, revenue level, and log-transformed site area. The revenue multiplier carries the largest absolute coefficient, followed by opportunity cost rate; site area exerts a positive but diminishing marginal effect consistent with economies of scale in logistics

512 and harvesting. Rotation length is positively associated with profit in low-opportunity-
513 cost settings but negatively so when foregone agricultural revenue is high, confirming the
514 rotation-length trade-off in the AFS-SCD formulation (Section 3).

Table 5: Approximate significance of smooth and tensor-product terms in the site-profit GAM (Gaussian, REML) and the scenario-level total AFS area GAM (Gamma, log link, REML). Logistics and establishment cost scalers enter the profit GAM as linear parametric terms (see Table 6); both appear as smooths in the area GAM, with an additional tensor-product interaction surface. R^2 and deviance explained: profit GAM $R^2 = 0.731$, dev. 73.1%; area GAM $R^2 = -0.388$, dev. 81.1%.

Model	Term	Eff. df	Ref. df	F	p
Site-profit GAM — smooth terms					
Profit GAM	s(site opp. cost)	3.750	3.955	4041.445	<0.001
Profit GAM	s(dist. site–hub, km)	3.962	3.999	380.365	<0.001
Profit GAM	s(avg. rotation, yr)	3.953	3.999	247.832	<0.001
Profit GAM	s(log AFS area)	2.983	3.000	117.448	<0.001
AFS area GAM — smooth terms					
Area GAM	s(opp. cost scaler)	1.906	1.991	794.190	<0.001
Area GAM	s(logistics cost scaler)	1.227	1.403	396.286	<0.001
Area GAM	s(establishment cost scaler)	1.003	1.006	262.428	<0.001
Area GAM	s(revenue scaler)	1.993	2.000	1783.132	<0.001

Table 6: Parametric (linear) coefficients in the site-profit GAM. The three scenario-level cost and revenue scalers are entered as linear terms; a unit increase in each scaler corresponds to the stated change in profit ($\text{€ ha}^{-1} \text{ yr}^{-1}$). The intercept is the expected profit at scaler = 0 and all smooth terms at their mean.

Predictor	$\hat{\beta}$	SE	t	p
<i>Intercept</i>	-115.76	2.68	-43.20	<0.001
Logistics cost scaler	-86.18	0.99	-86.89	<0.001
Establishment cost scaler	-55.89	0.96	-58.00	<0.001
Revenue scaler	544.15	2.50	217.68	<0.001

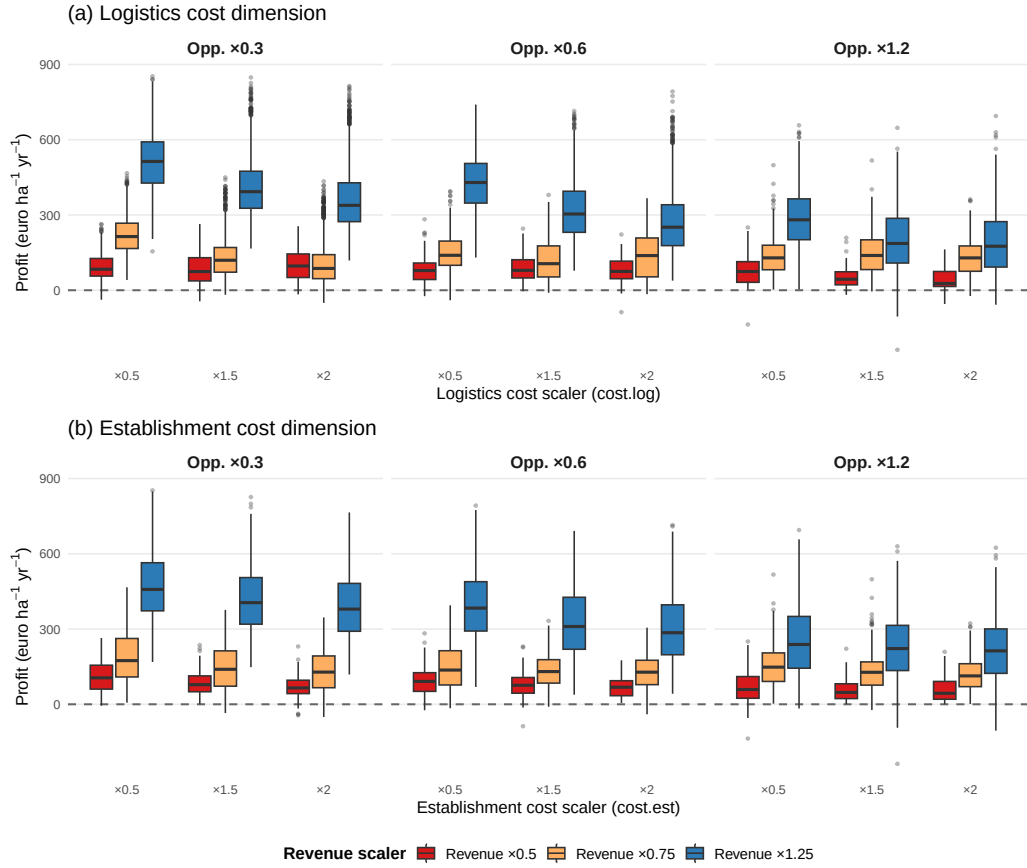


Figure 13: Distribution of site-level profit ($\text{€ ha}^{-1} \text{yr}^{-1}$) across scenario parameter levels. Each panel row shows one cost dimension (logistics cost scaler cost.log , top; establishment cost scaler cost.est , bottom); columns show opportunity cost level (opp , relative to baseline $150 \text{ € ha}^{-1} \text{yr}^{-1}$). Fill colour encodes the revenue scaler. Each box summarises all active sites across all 10 stochastic replications of that parameter combination. The dashed horizontal line marks zero profit.

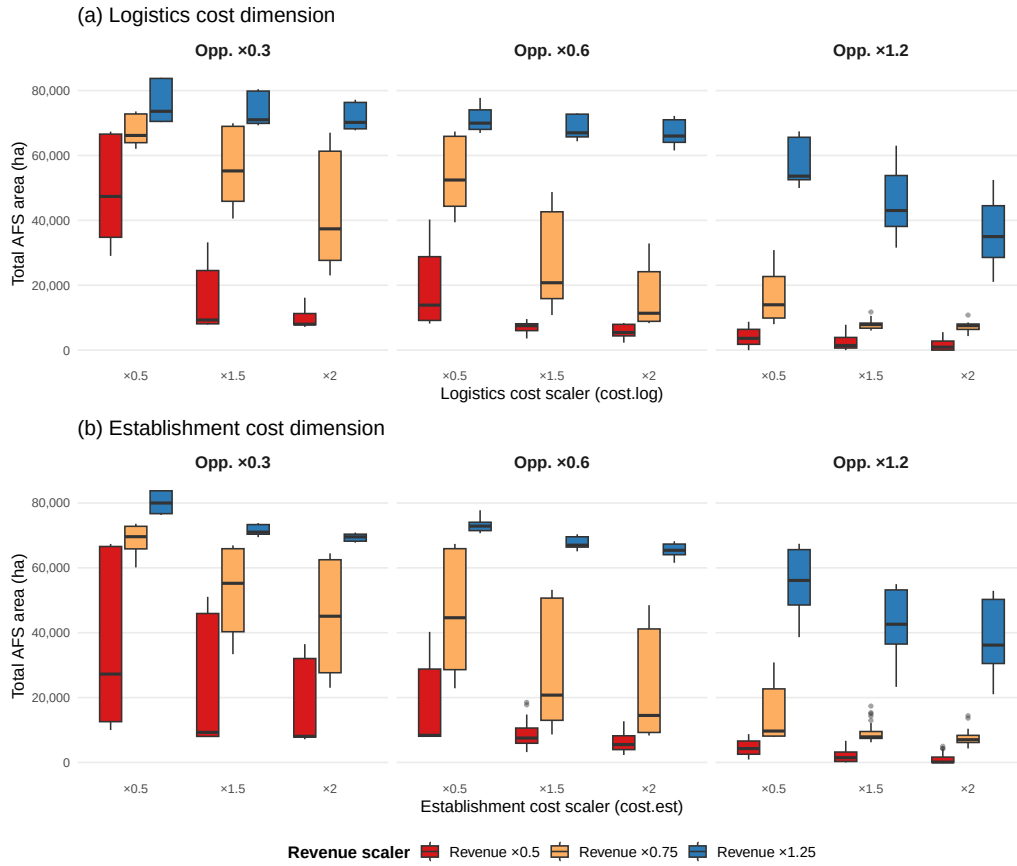


Figure 14: Distribution of total AFS area dedicated per scenario replicate (ha) across parameter levels. Each panel row shows one cost dimension (logistics cost scaler *cost.log*, top; establishment cost scaler *cost.est*, bottom); columns show opportunity cost level. Fill colour encodes the revenue scaler. Each box summarises the 10 stochastic replications of that parameter combination. Higher values indicate broader spatial activation of the AFS potential.

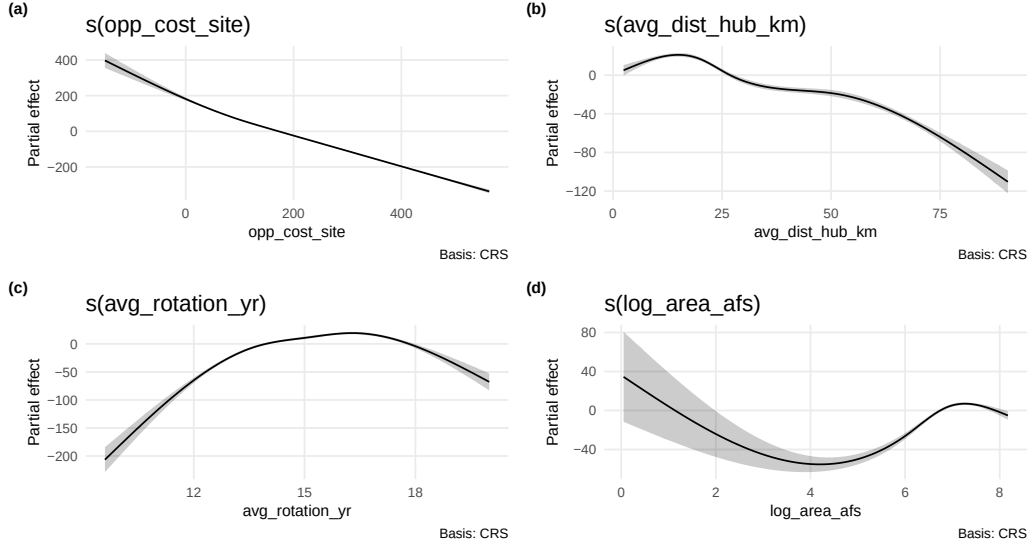


Figure 15: Partial effects (centred smooths) of the four non-linear terms in the site-profit GAM: site-level opportunity cost, hub distance, average harvest rotation length, and log-transformed AFS strip area. Shaded bands: 95 % simultaneous credible intervals. Deviance explained: 73.1 %; adjusted $R^2 = 0.731$.

6. Conclusion

This paper has developed and applied the AFS-SCD model, a MILP formulation for the integrated design and operation of agroforestry biomass supply chains. The model endogenises stand age dynamics through binary arc-based age-lag variables, couples component-specific Gompertz yield coefficients for stem, branch, and residue fractions with multi-echelon logistics decisions, and embeds a product quality cascade that allocates biomass grades to chemical, pulp, and energy markets within a unified optimisation framework. The formulation is, to the best of our knowledge, the first supply chain model to simultaneously optimise AFS establishment timing, consecutive harvest scheduling under rotation-age constraints, and product-grade allocation across spatially heterogeneous consumer markets.

The case study for the Saxony-Anhalt tri-state region demonstrates the practical relevance of the approach. The optimal supply chain concentrates AFS establishments in the Elbe and Saale floodplains, where high biomass productivity and proximity to large pulp mills create the most favourable economics. The sensitivity analysis confirms that

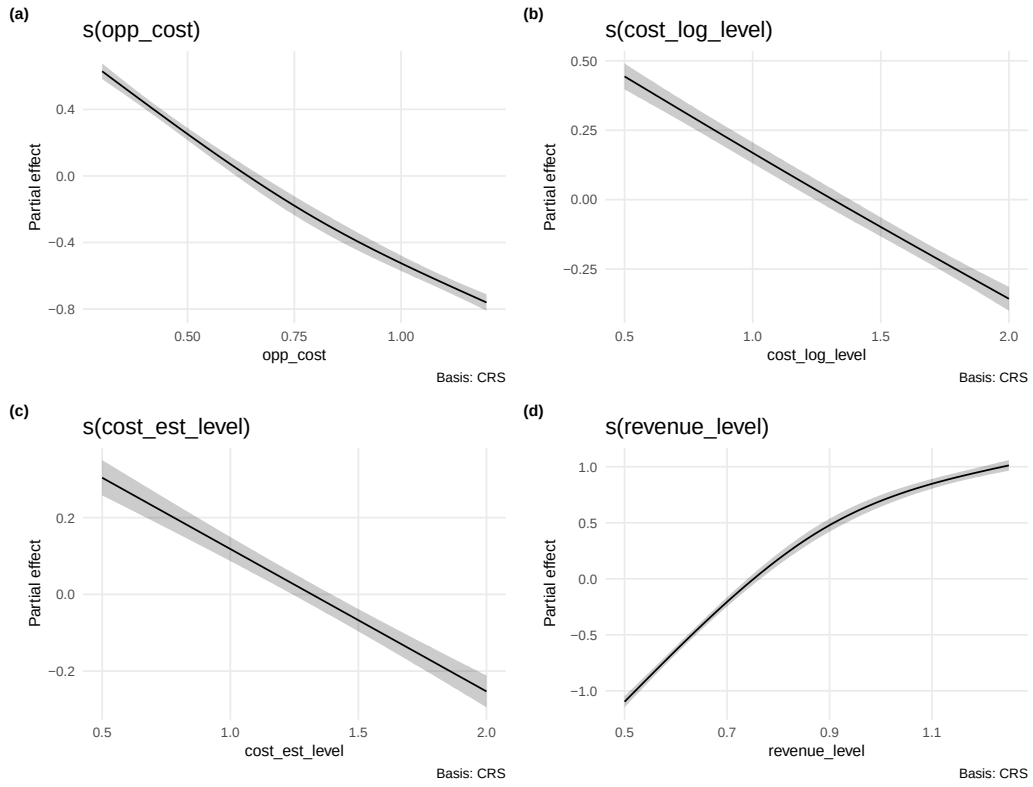


Figure 16: Partial effects of scenario-parameter smooths in the scenario-level total AFS area GAM (Gamma family, log link). Panel (e) shows the tensor-product interaction surface of logistics and establishment cost scalars; contours indicate predicted total AFS area on the response scale (ha). Shaded bands in panels (a)–(d): 95% simultaneous credible intervals. Deviance explained: 81.1%; adjusted $R^2 = -0.388$.

profitability depends primarily on achievable gate prices for high-quality (P1) biomass grades: scenarios combining reduced revenue with high opportunity costs yield negative net profits, whereas scenarios with full or elevated prices remain viable under cost pressure. Opportunity cost is the dominant site-level discriminator, with parcels exceeding approximately $300 \text{ € ha}^{-1} \text{ yr}^{-1}$ rarely selected in the optimal solution. These findings underscore the importance of product cascading as a strategic lever for AFS viability: maximising the share of stem biomass directed to high-value chemical and pulp applications substantially improves profitability compared to energy-only offtake strategies.

Methodologically, the AFS-SCD model bridges the gap between stand-level growth modelling and regional supply chain optimisation identified in the literature (Zahraee et al., 2020; Lamers et al., 2011). The age-lag constraint structure generalises to other perennial biomass crops with periodic coppice cycles, such as willow or black locust, and to long-rotation silvoarable systems in which rotation length is a strategic variable. The model is extensible to stochastic settings by replacing deterministic yield coefficients with scenario-weighted counterparts or by incorporating robust optimisation over yield uncertainty intervals (Sharma et al., 2013).

Several limitations point to directions for future research. First, consumer demand is treated as exogenous and time-invariant; endogenising demand responses to biomass price changes would require a partial-equilibrium extension. Second, the case study uses a single allometric scenario calibrated to Central European poplar; propagating Gompertz parameter uncertainty through to supply chain outcomes would strengthen the robustness analysis. Third, environmental co-benefits of AFS — including carbon sequestration, nitrate leaching reduction, and biodiversity enhancement — are not monetised in the current objective. Integrating life cycle assessment indicators as additional objectives or constraints would support a more comprehensive sustainability evaluation, consistent with the cleaner production framing of this work.

7. Appendix

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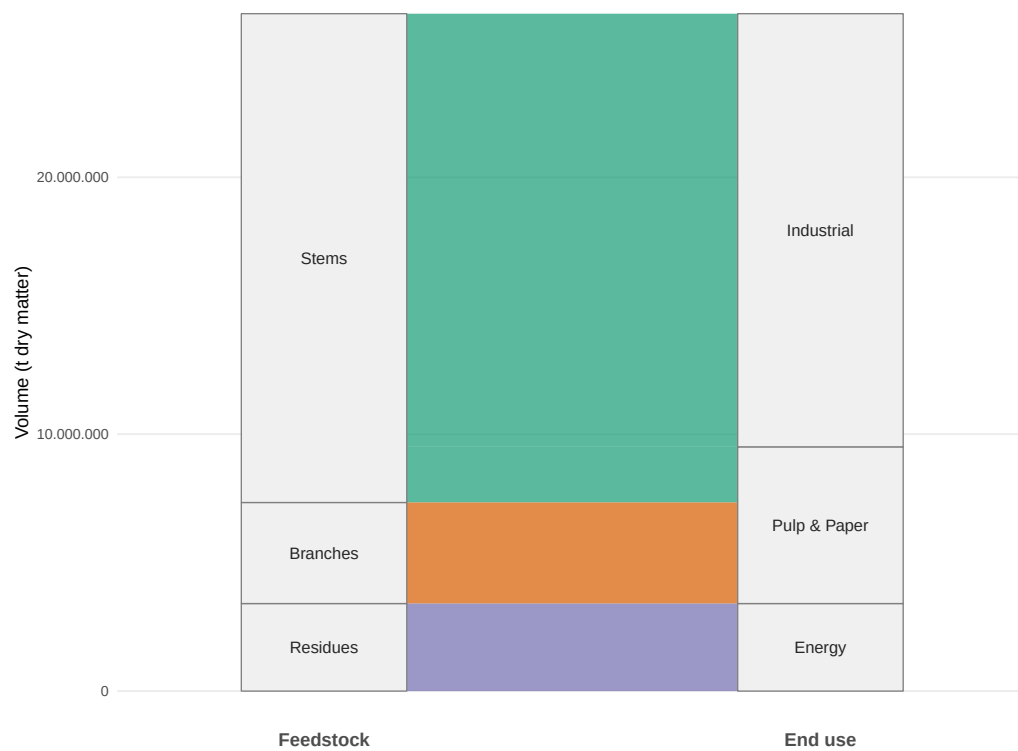


Figure 17: Total production quantity of biomass types and its industrial usage over planning horizon (in tons).

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Table 7: Consumer demand quantities and gate prices by product grade. P1: chemical/pulp; P2: sawlog/pulpwood; P3: energy/pellets. Demands in kt wood input per year; prices in €/t wood input at the factory gate.

Consumer	Annual demand			Gate price		
	P1 (kt/yr)	P2 (kt/yr)	P3 (kt/yr)	P1 (€/t)	P2 (€/t)	P3 (€/t)
consumer 1	300.0	–	–	50	–	–
consumer 2	100.0	–	–	60	–	–
consumer 3	–	300	50.0	–	37.5	24
consumer 4	200.0	–	–	47.5	–	–
consumer 5	–	30	5.0	–	35	21
consumer 6	–	15	5.0	–	32.5	21
consumer 7	–	10	3.0	–	30	18
consumer 8	–	5	8.0	–	27.5	18
consumer 9	2.0	50	20.0	40	35	22.8
consumer 10	–	–	1.7	–	–	21
consumer 11	–	–	2.9	–	–	21
consumer 12	–	–	2.5	–	–	21
consumer 13	–	–	2.3	–	–	21
consumer 14	–	–	1.9	–	–	21
consumer 15	–	–	2.7	–	–	21
consumer 16	–	–	1.7	–	–	21
consumer 17	–	–	1.9	–	–	21
consumer 18	–	–	1.6	–	–	21
consumer 19	–	–	1.5	–	–	21
consumer 20	–	–	1.5	–	–	21
consumer 21	–	–	1.8	–	–	21
consumer 22	–	–	3.5	–	–	21
consumer 23	–	–	1.6	–	–	21
consumer 24	–	–	2.1	–	–	21
consumer 25	–	–	4.5	–	–	21
consumer 26	–	–	1.8	–	–	21
consumer 27	–	–	3.2	–	–	21
consumer 28	18.5	–	–	27.5	–	–